

- Dineswara kumar
- BT24CSA062 1) Generate a checkerboard matrix of size 8×8 Hint: Use slicing with steps.

o/p:

- `[[0 1 0 1 0 1 0 1]`
- `[1 0 1 0 1 0 1 0]`
- `[0 1 0 1 0 1 0 1]`
- `[1 0 1 0 1 0 1 0]`
- `[0 1 0 1 0 1 0 1]`
- `[1 0 1 0 1 0 1 0]`
- `[0 1 0 1 0 1 0 1]`
- `[1 0 1 0 1 0 1 0]]`

2) Multiply two large matrices efficiently and check execution time

3) Compute eigenvalues & eigenvectors of a symmetric matrix

4) Solve a system of linear equations Example: $2x + y = 5$ and $x - y = 1$

5) Generate random numbers from a multivariate normal distribution

6) Student Marks Data Analysis using NumPy

Stores marks of students in multiple subjects

Calculates average, highest, lowest, and standard deviation

Finds the top performer and subject-wise analysis

```
1.
import numpy as np

checkerboard = np.zeros((8, 8), dtype=int)
checkerboard[1::2, ::2] = 1
checkerboard[:, 1::2] = 1
```

```
print(checkerboard)
```

```
[[0 1 0 1 0 1 0 1]
 [1 0 1 0 1 0 1 0]
 [0 1 0 1 0 1 0 1]
 [1 0 1 0 1 0 1 0]
 [0 1 0 1 0 1 0 1]
 [1 0 1 0 1 0 1 0]
 [0 1 0 1 0 1 0 1]
 [1 0 1 0 1 0 1 0]]
```

```
2
import numpy as np
import time
```

```
size = 10
matrix1 = np.random.rand(size, size)
matrix2 = np.random.rand(size, size)
s_time=time.time()
result_matrix = np.dot(matrix1, matrix2)
e_time=time.time()

print(f"Execution time ", (e_time-s_time ))
```

```
Execution time  0.00033664703369140625
```

```
4
import numpy as np

A = np.array([[2, 1], [1, -1]])
b = np.array([5, 1])
solution = np.linalg.solve(A, b)

print(f" x = {solution[0]}, y = {solution[1]}")
```

```
x = 2.0, y = 1.0
```

```
# 6
average_marks = np.mean(marks)
highest_mark = np.max(marks)
lowest_mark = np.min(marks)
std_dev = np.std(marks)

print(f"\nAverage Marks: {average_marks:.2f}")
print(f"Highest Mark: {highest_mark}")
print(f"Lowest Mark: {lowest_mark}")
print(f"Standard Deviation: {std_dev:.2f}")

total_marks_per_student = np.sum(marks, axis=1)
topper_index = np.argmax(total_marks_per_student)
topper_score = total_marks_per_student[topper_index]

print(f"\nTop Performer is Student {topper_index + 1} with a total
score of {topper_score}")
```